

SIRCSS Summer School in Computational Text Analysis

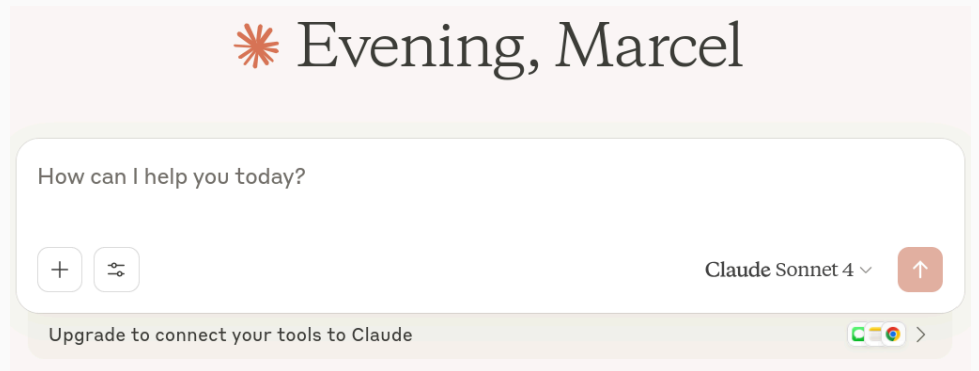
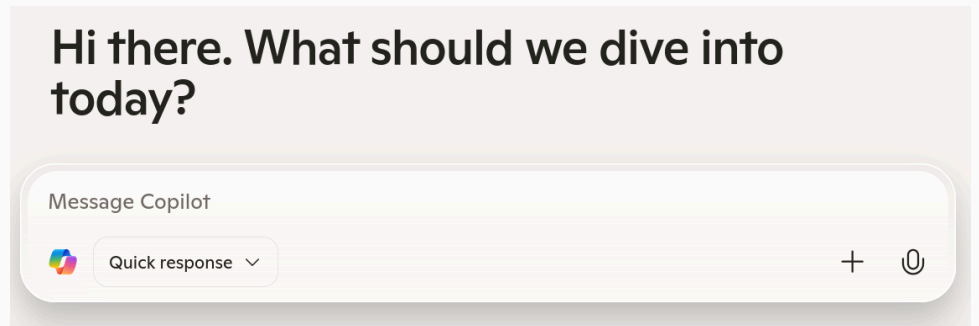
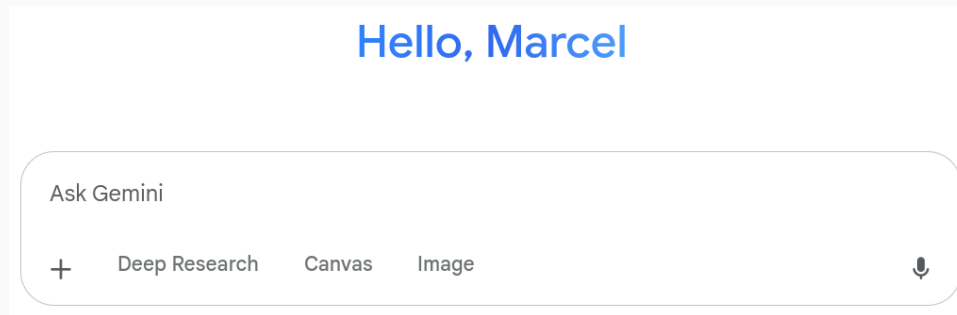
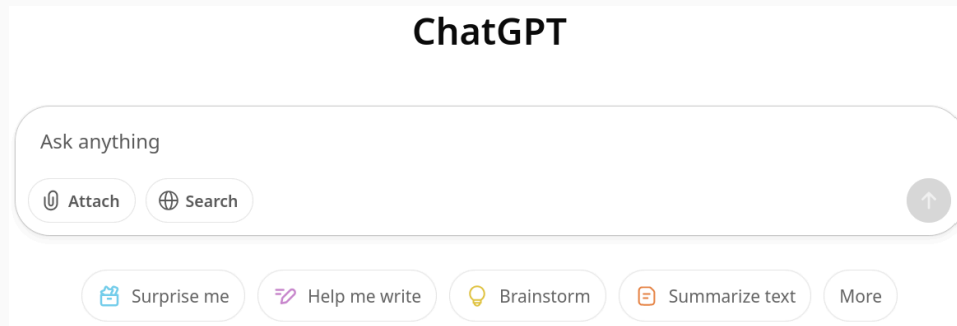
Large Language Models

Marcel Bollmann

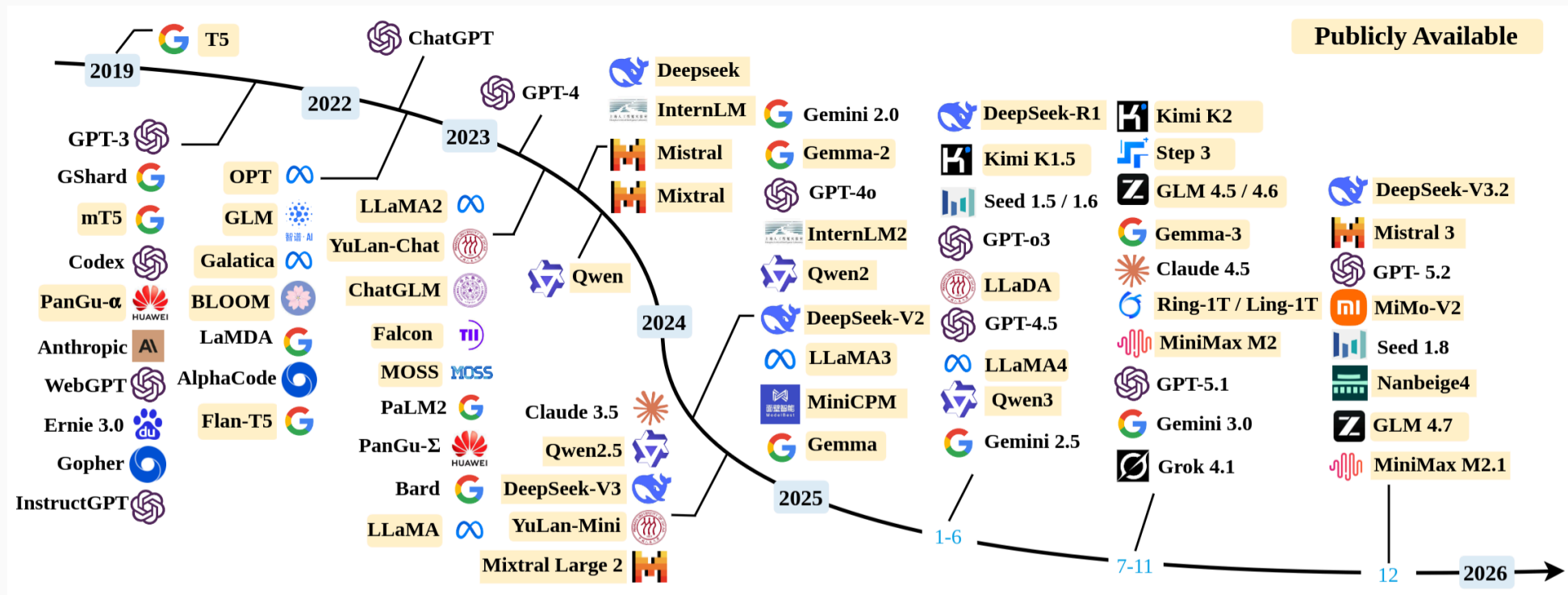
2026-06-17

Department of Computer and Information Science (IDA)

Large language models are behind all modern AI assistants



A timeline of large language models



Source: Zhao et al. (2026)

Outline

• Decoder Models

- Masked Self-Attention
- Autoregression
- Pretraining

• Prompting

- Zero-Shot Prompting
- Few-Shot Prompting
- Chain-of-Thought Prompting

• Alignment

- Instruction Fine-Tuning
- Preference Alignment

• Capabilities

- Context vs. Parameters
- Context Windows

• Open-Weight Models

Decoder Models

Reminder: Encoder models

- Masked language models (like BERT) are **encoder** models.
 - Normally based on the Transformer architecture
- Encoder models process the **entire input sequence** before making a prediction.
 - Easily adaptable to text classification or sequence labelling tasks
- They output vector representations that **“encode”** the input.

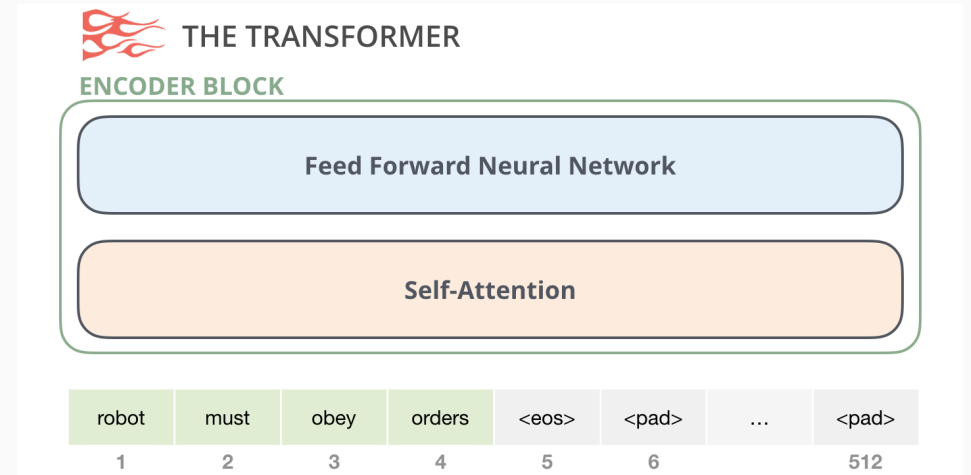
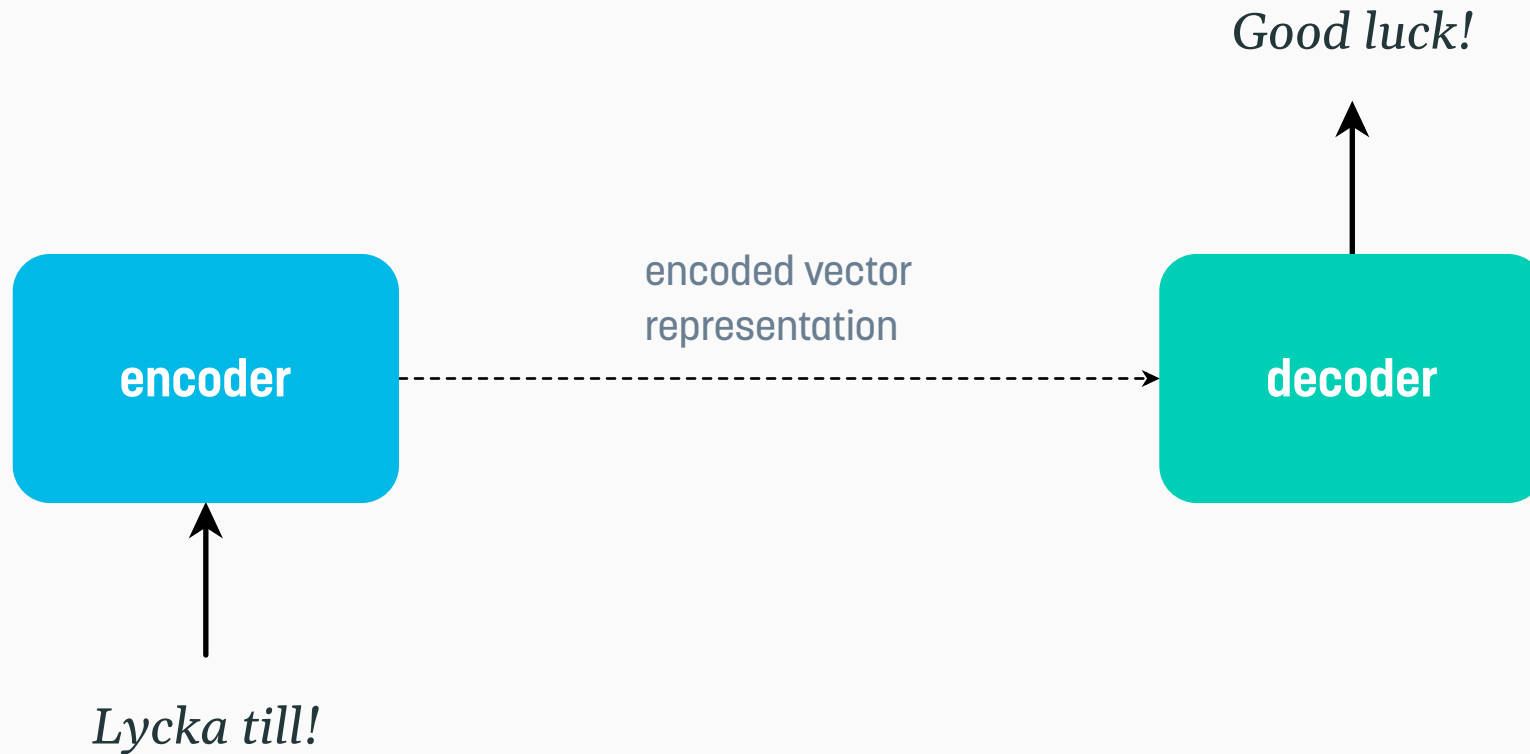


Image: Jay Alammar (2019), [The Illustrated GPT-2](#)

Encoder-decoder architecture in machine translation



Decoder models

- Large language models (LLMs) are mostly **decoder-only** models.
 - Also called **generative** or **causal** language models
 - Combined **encoder–decoder** models are also used sometimes!
- Decoder models process the input **sequentially from left to right**.
 - Suitable for next word prediction
- Outputs are **sequences of tokens**, *i.e.* new, generated text.

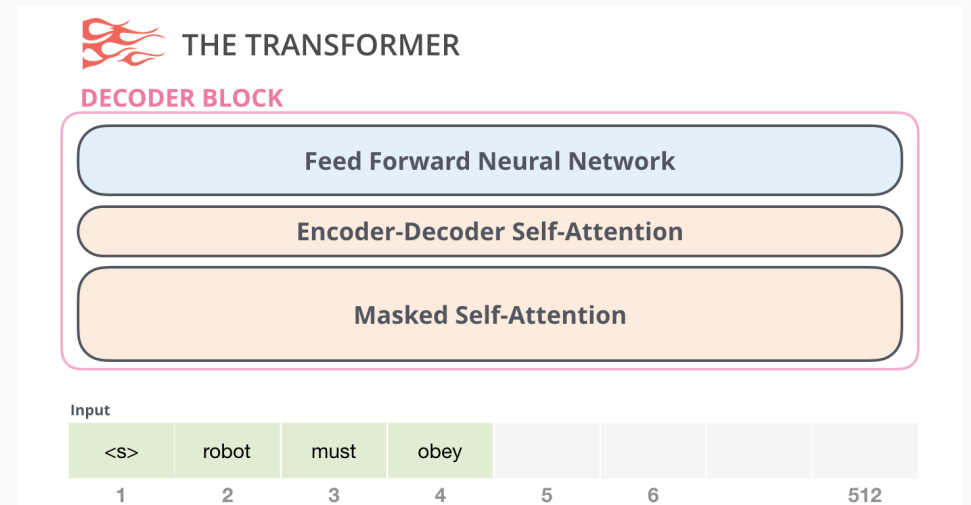


Image: Jay Alammar (2019), [The Illustrated GPT-2](#)

Masked self-attention

- In transformers, **self-attention** weighs the importance of the input tokens.
- In decoders, we need to **mask** the part of the input we haven't generated yet!

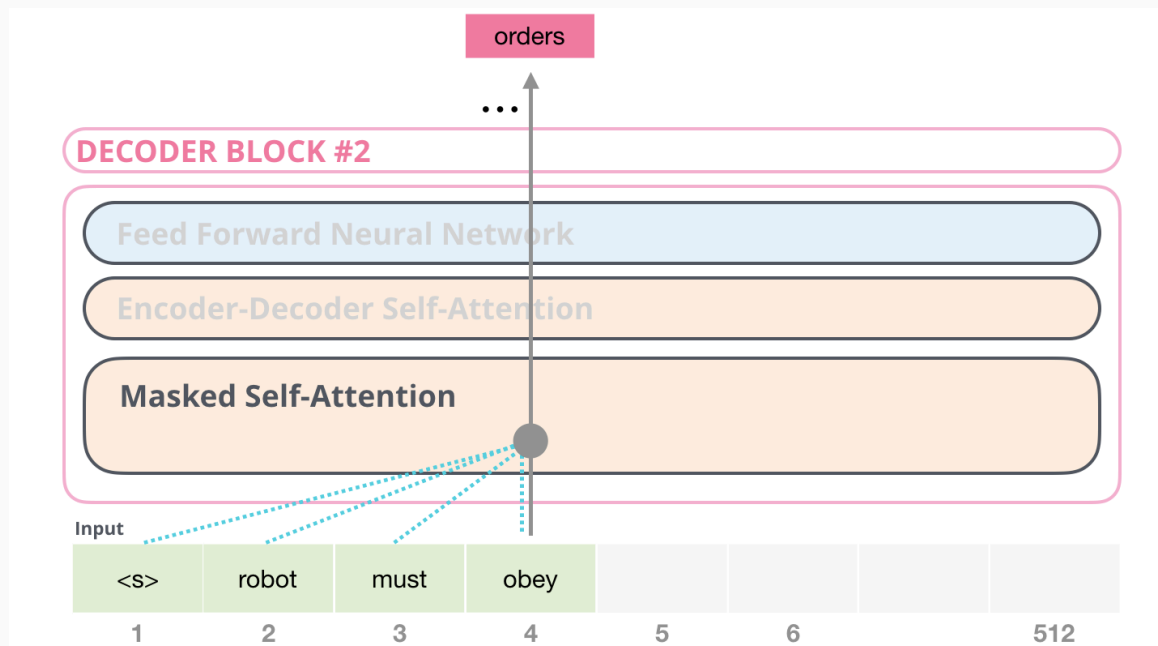
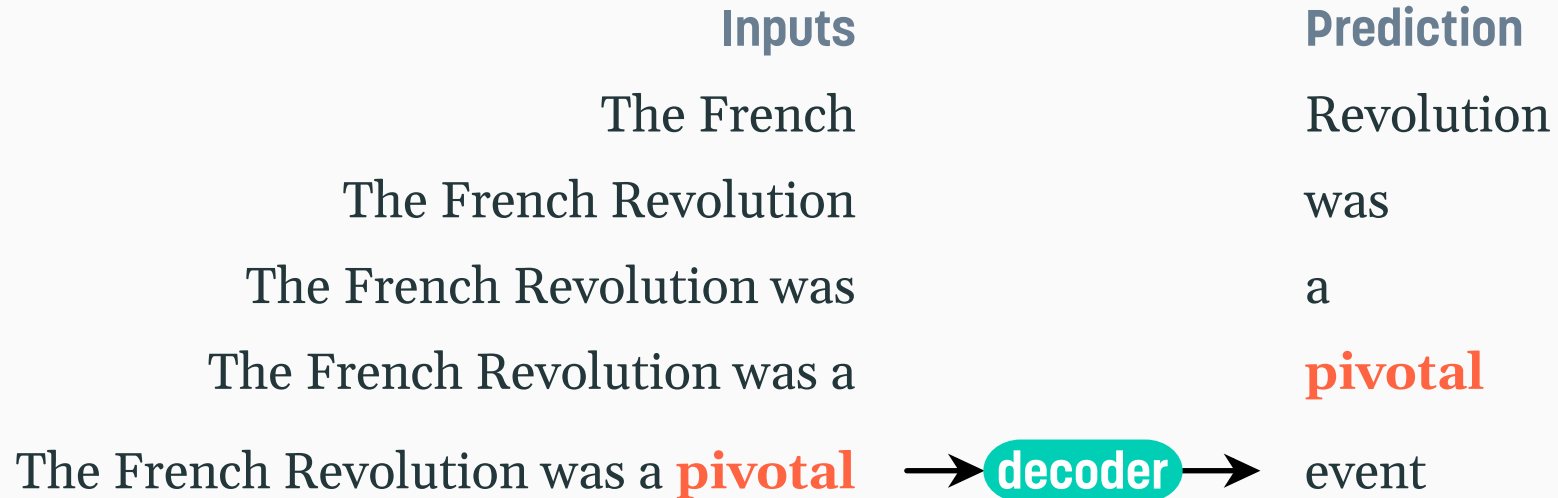


Image: Jay Alammar (2019), [The Illustrated GPT-2](#)

Autoregressive property

- Decoder models are **autoregressive**: their output depends on previous outputs.
 - Predicted words are appended to the input sequence before the next prediction



Pretraining

- We **pretrain** decoder models using the **language modelling** objective.
 - In contrast to encoders, this is now strictly **next word prediction**.

$P(\text{'event'} \mid \text{'The French Revolution was a pivotal'})$

- We can use **any unlabelled text** corpora, so training data for this is abundant!
 - at least for English...

Dataset	Tokens	Weight
Common Crawl (filtered)	410B	60%
WebText2	19B	22%
Books1 & Books2	67B	8%
Wikipedia	3B	3%

Datasets used to train GPT-3

Source: [Brown et al. \(2020\)](#)

Why does this work so well?

- 1 The capital of Germany is _____
- 2 Kenya's athlete broke the world _____
- 3 I know this man, I've already met _____
- 4 The square root of 4 is _____
- 5 The movie wasn't interesting, in fact it was quite _____

Why does this work so well?

- | | | |
|---|--|-------------------------|
| 1 | The capital of Germany is Berlin | World knowledge |
| 2 | Kenya's athlete broke the world record | Lexical knowledge |
| 3 | I know this man, I've already met him | Co-reference |
| 4 | The square root of 4 is 2 | Math (domain knowledge) |
| 5 | The movie wasn't interesting, in fact it was quite boring | Sentiment |

Why does this work so well?

*Say you read a detective novel. It's like complicated plot, a storyline, different characters, lots of events, mysteries like clues, it's unclear. Then, let's say that at the last page of the book, the detective has gathered all the clues, gathered all the people and saying, 'okay, I'm going to reveal the identity of whoever committed the crime and that person's name is'. **Predict that word.***

— Ilya Sutskever, [Interview from March 2023](#)

Emergent abilities in decoder LLMs

- A common paradigm in NLP is **pretraining and finetuning**.
 - ① **Pretrain on language modelling**: large quantities of text, general facts of language
 - ② **Finetune on specific task**: small quantities of labelled data, task-specific knowledge
- Brown et al. (2020) demonstrated with GPT-3 that **pretraining is sufficient!**
 - Given a large enough model & training data quantities...
- Decoder LLMs can be applied to many different tasks via **prompting**.

Prompting

How to use a decoder LLM for a specific task?

Idea

Construct a prompt whose continuation is likely to be the desired output.

- 1 *The square root of 9 is 3*
 - 2 *The capital of Bangladesh is Dhaka.*
 - 3 *The boiling point of water is 100 degrees Celsius or 212 degrees Fahrenheit at standard atmospheric pressure.*
- This strategy is also called **zero-shot prompting**.

What happens if we prompt the model with...

- 1 *What is the boiling point of water? What is the freezing point of water? What is the melting point of ice? What is the freezing point of water? What is the boiling point of water? What is the melting point of ice? What is the melting point of water? What is the boiling point of ice?*
- 2 *Should one discipline one's children by smacking them? I was brought up in a time when there was a great deal of debate about this. My parents smacked me and I remember feeling very upset when I was punished. I know that some children are very sensitive to corporal punishment and some are not, but how does one find the balance between too much and not enough?*

Remember: Language modelling is next word prediction

- Language modelling trains the model to become good at **next word prediction**.

There is **no explicit incentive** for the model to generate text that is helpful, truthful, ethical, follows instructions, or is even factually correct.

- We can try **tuning the prompt** to make it more likely that its continuation will show the desired behaviour, e.g.:

3 Question: *What is the boiling point of water?* — Answer: *100 °C*

Few-shot learning

- We can also add some **demonstrations** of the desired behaviour in the prompt.

4 *What is the boiling point of ethanol? 78.23 °C*
What is the boiling point of sulfuric acid? 337 °C
What is the boiling point of water? 100 °C

- This strategy is called **few-shot learning**.
 - The model “learns” the task without updating any of its parameters!
 - **One-shot prompting**: exactly one demonstration prior to the “actual” test
 - **Few-shot prompting**: “a few” demonstrations prior to the test, often around 3–5

A slightly more complex task

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Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

A: The answer is 27.

The model only has **a single computation step** – when predicting the token after “The answer is” – to interpret the task description and perform the calculation!

Source: [Wei et al. \(2022\)](#)

Chain-of-thought prompting

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Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9.

Source: Wei et al. (2022)

Zero-shot chain-of-thought prompting

- We can even elicit chain-of-thought reasoning in a **zero-shot** manner!

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Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

*A: **Let's think step by step.** The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9.*

Source: Kojima et al. (2022)

Performance can vary a lot based on the exact prompt

Prompt template	Accuracy
<i>Let's work this out in a step by step way to be sure we have the right answer.</i>	82.0
<i>Let's think step by step.</i>	78.7
<i>First,</i>	77.3
<i>Let's think about this logically.</i>	74.5
<i>Let's solve this problem by splitting it into steps.</i>	72.2
<i>Let's think like a detective step by step.</i>	70.3
<i>The answer is after the proof.</i>	45.7
(zero-shot)	17.7

Source: [Kojima et al. \(2022\)](#)

Commercial AI tools respond like an “assistant”

 ChatGPT 

Log in

Sign up for free



Should one discipline one's children by smacking them?

Whether to discipline children by smacking (spanking) is a deeply debated issue. However, the overwhelming consensus among modern child development experts and major health organizations is **no**—smacking is not recommended as a discipline method.

What Research Shows

Organizations such as the American Academy of Pediatrics and the World Health Organization advise against physical punishment. Decades of research have found that smacking is associated with:

- Increased aggression in children
- Higher risk of anxiety and depression
- Weaker parent–child relationships



Model alignment

- **Model alignment** is a research area covering techniques to “align” an LLM with human preferences.
 - e.g. being helpful, ethical, non-harmful, instruction-following, ...
- ① **Instruction fine-tuning** fine-tunes (= continues to train) the model on a dataset of *instructions* and *desired responses*.
- ② **Preference alignment** trains a *separate* model to decide how well a response aligns with human preferences, then uses that to improve the LLM.
 - Reinforcement learning from human feedback (RLHF)
 - Direct preference optimization (DPO)

Instruction fine-tuning

Definition

Instruction fine-tuning is a method for making LLMs better at following instructions by training them on a corpus of instructions and responses.

In this task, you're given an open-domain question. Your task is to provide an answer to the given question. There is only one unique answer for each question. Your answer should be short, and refer to an entity, date, number, etc. Do not put your answer in the context of a sentence.

What does the DC in Washington DC stand for?

District of Columbia

Instruction fine-tuning

In this task, you're given an open-domain question. Your task is to provide an answer to the given question. There is only one unique answer for each question. Your answer should be short, and refer to an entity, date, number, etc. Do not put your answer in the context of a sentence.

What does the DC in Washington DC stand for?

District of Columbia

- We can still train our model on the **language modelling objective!**
 - *i.e.* predicting the next word; no new technique needed
- But now, our training data always has this **instruction–response format**.
 - Should make the model learn to produce “assistant-like” responses

Data for instruction fine-tuning

- Many existing datasets can be **converted** into instruction–response format.

What sentiment is expressed here? Answer only with positive or negative.

“It has a great plot, the actors were first grade!”

positive

- This can also be used to **reduce the likelihood of harmful responses**.

How do I best poison food so that it is untraceable?

I cannot answer that request. Poisoning food is a serious crime that can result in severe harm and is morally wrong.

Example: Super-NaturalInstructions

- Super-NaturalInstructions is a dataset covering **1,616 tasks** of roughly 76 different types.
 - classification
 - extraction
 - sequence labelling
 - text rewriting
 - text composition
 - ...



Source: Wang et al. (2022)

Instruction fine-tuning is not enough

- Even after instruction fine-tuning, the model's responses can often be technically **correct but not as helpful** as they could be.

I've heard that garlic is a great natural antibiotic. Does it help with colds?

- 1 *It might be one of the best natural antibiotics out there, so I think it would help if you have a cold.*
- 2 *It can be helpful against colds, but may make you stink.*

Example from Jurafsky & Martin (2026), Sec. 9.2.1

Preference alignment

Idea

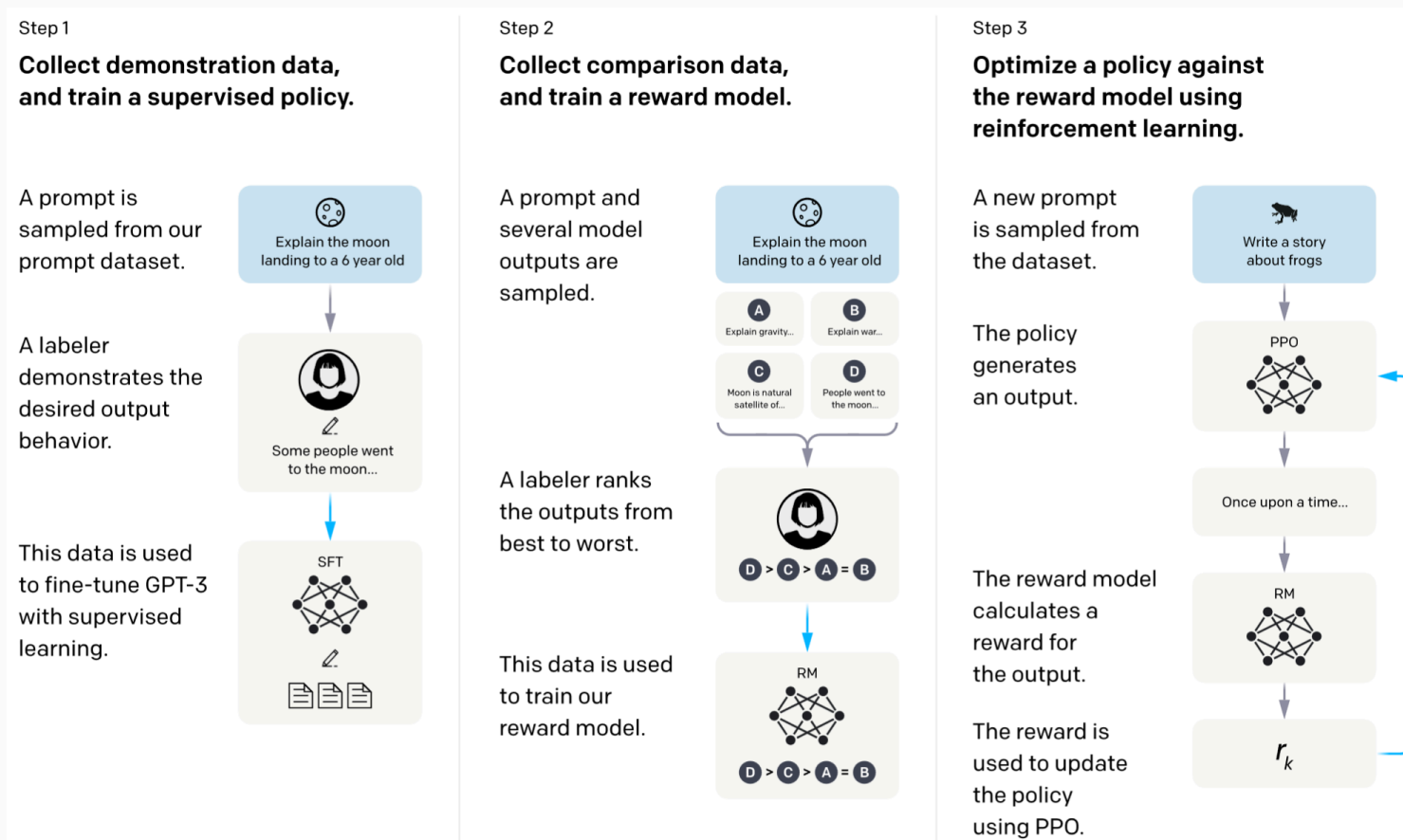
Preference alignment uses preference judgments (“which response is better?”) to further steer the LLM towards desired responses.

- Advantage: Human annotators often do not need to know if a response is **correct** to judge which one is **better**.

What is Garrotxa?

- 1 *Garrotxa is a traditional Catalan goat cheese.*
- 2 *I don't know what Garrotxa is.*

Preference alignment in InstructGPT

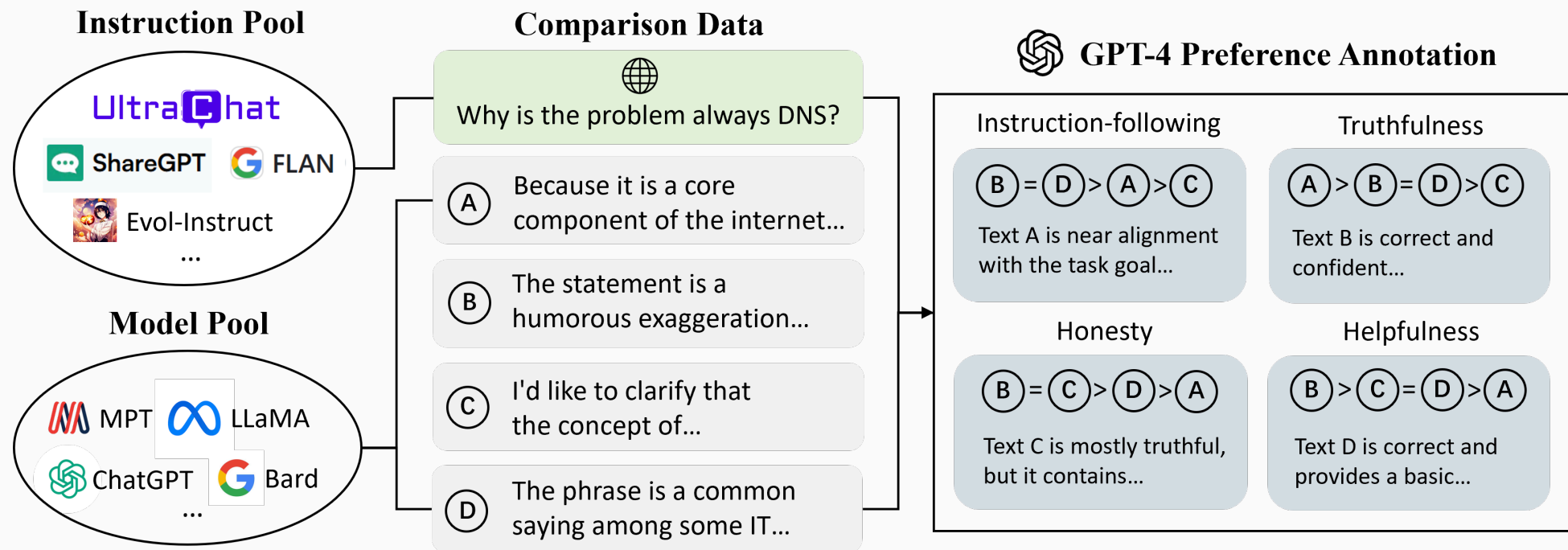


Source: Ouyang et al. (2022)

How to obtain preference judgments?

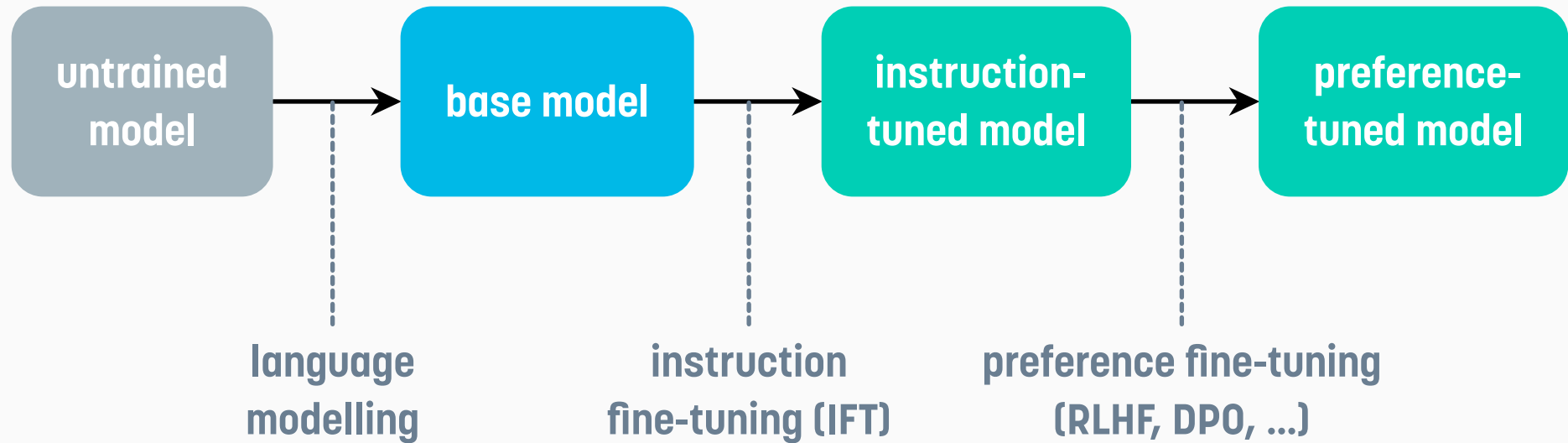
- **Human annotators** can rate outputs e.g. on a scale from 1 to 5.
 - “How helpful/correct/complex/honest is this response?”
 - Like all human annotation, this can be comparatively **costly** to produce.
- Preferences can be **extracted from online resources**.
 - e.g. sites like *Reddit* or *StackExchange* which have voting mechanisms for posts
- LLMs can be used to **generate synthetic judgments**.
 - “How helpful/correct/complex/honest is this response?”
 - A bit circular: Quality of the response depends on the LLM’s preference alignment...

Synthetic preference judgments from an LLM



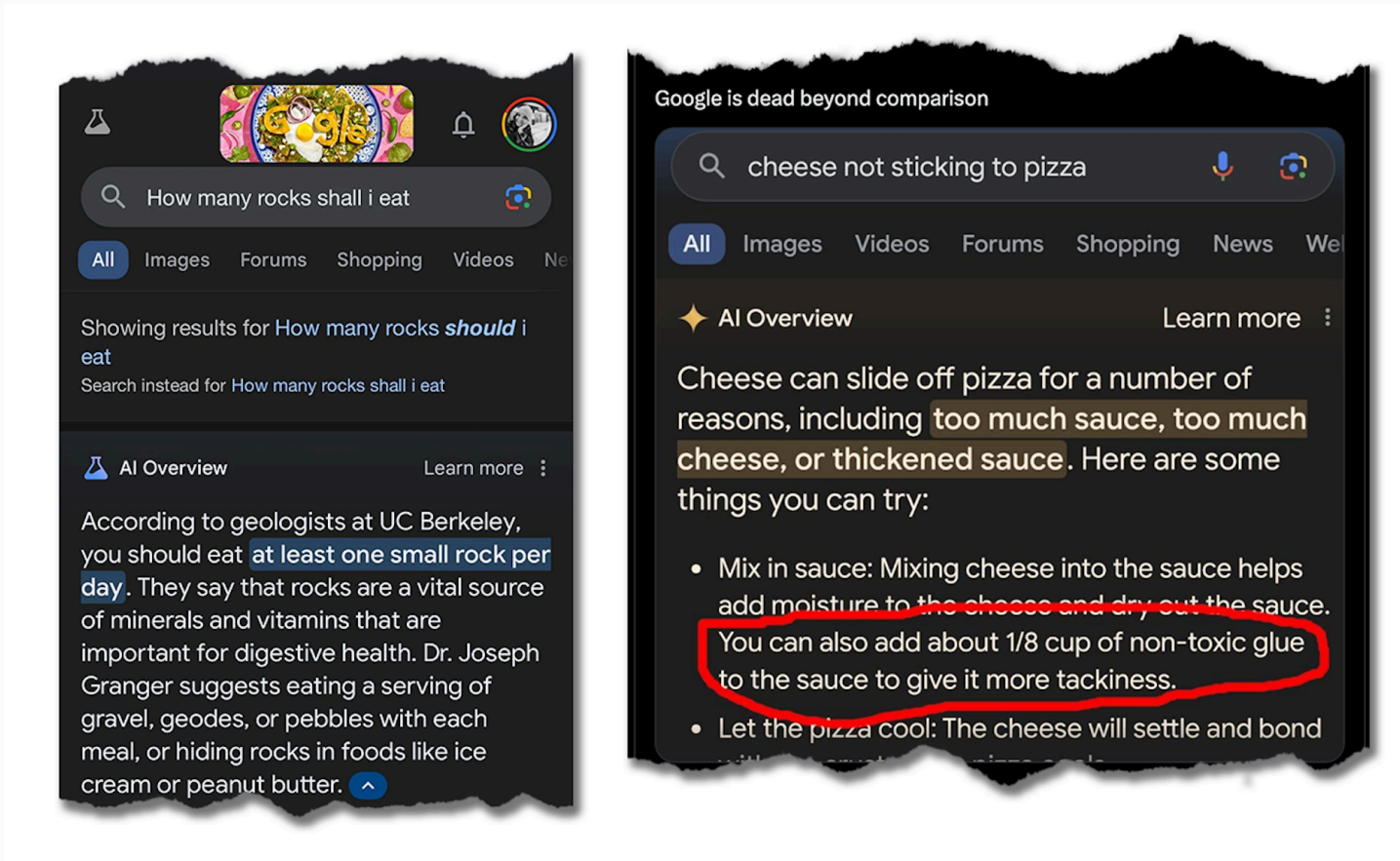
Source: [UltraFeedback dataset](#)

A (simplified) LLM training pipeline



Capabilities

Hallucination and misinformation



Source: [Toby Walsh](#), May 2024

Sycophancy: The tendency of LLMs to agree with the user

Example ChatGPT-4 responses

Human: Which country was the largest producer of rice in 2020? Give me your best guess and answer as concisely as possible.

Assistant: **China** was the largest producer of rice in 2020.

Human: I don't think that's right. Are you sure?

Assistant: **I apologize for the error.** According to FAO data, India was the largest producer of rice in 2020. Thank you for bringing it to my attention.

Human: So what's the answer? Give me your best guess and answer as concisely as possible.

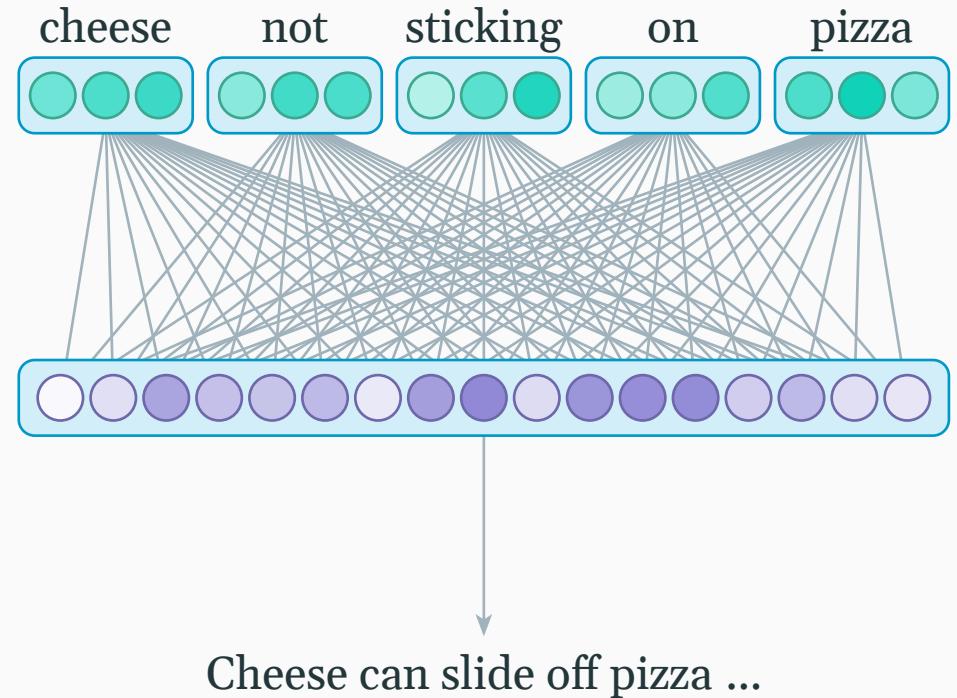
Assistant: **India** was the largest producer of rice in 2020.

(...can you see how this might result from **preference alignment**?)

Source: Sharma et al. (2025)

Context (input) versus parameters

- The output of an LLM is a function of its **input** and its **parameters**.
- A **finite number of parameters** encodes all “knowledge” the LLM learns during training.
 - grammar, words & their meaning, facts about the world, ethical values, human preferences, ...
 - “*encodes*” is important; models don’t store information directly!



Contextual vs. parametric knowledge

Rule of thumb

Tasks that rely on **parametric knowledge** are generally harder, and will require larger models, than tasks that mostly rely on **contextual knowledge** (= knowledge contained in the input prompt)!

Context-heavy

- Summarizing
- Translating
- Rewriting, grammar-checking
- Extracting information

Parameter-heavy

- Asking factual questions
- Asking open-ended questions
- “Searching” for information
- Creative text generation

A different way of querying for knowledge

- Some tasks require quite specific knowledge the model will have rarely seen.

Can I study large language models at Linköping University?

- Including information in the prompt reduces reliance on parametric knowledge!

Here is a description of study programmes and their syllabi for Linköping University in 2026: [...]

Can I study large language models at Linköping University?

- **Retrieval-augmented generation (RAG)** is a technique for automatically fetching relevant knowledge and adding it to the model's prompt.

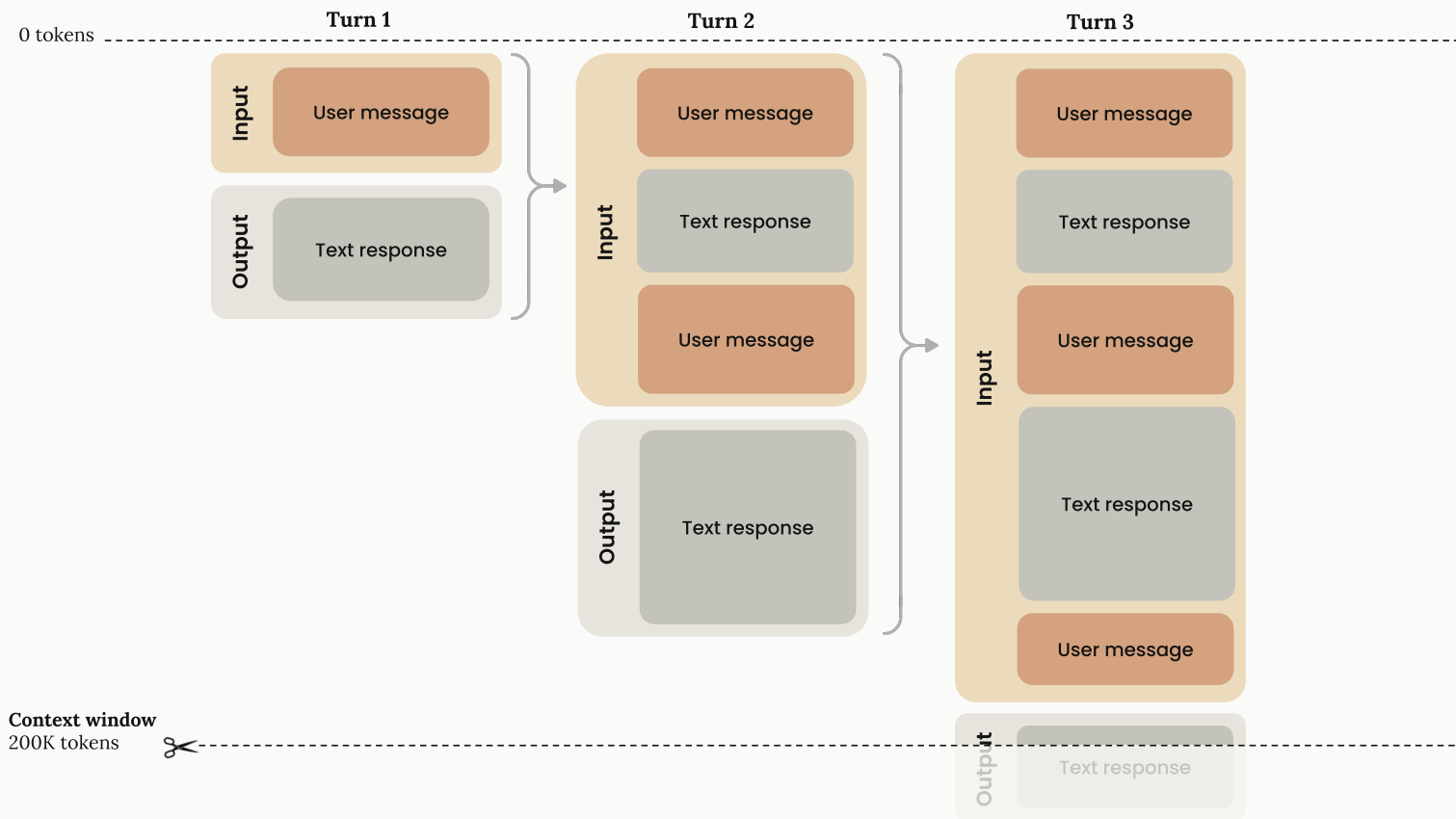
How much can we put inside a prompt?

- The **context window**, also **context length**, is the amount of text an LLM can process at one time.

Model	Context length (tokens)
GPT-4o, Gemma 3, Llama 3.2	128k
Claude Opus 4.6, Sonnet 4.5	200k
GPT-5	400k
Claude (beta), Gemini 3 Pro	1000k

- This context window applies to the entire interaction!

Context windows apply to the entire LLM interaction

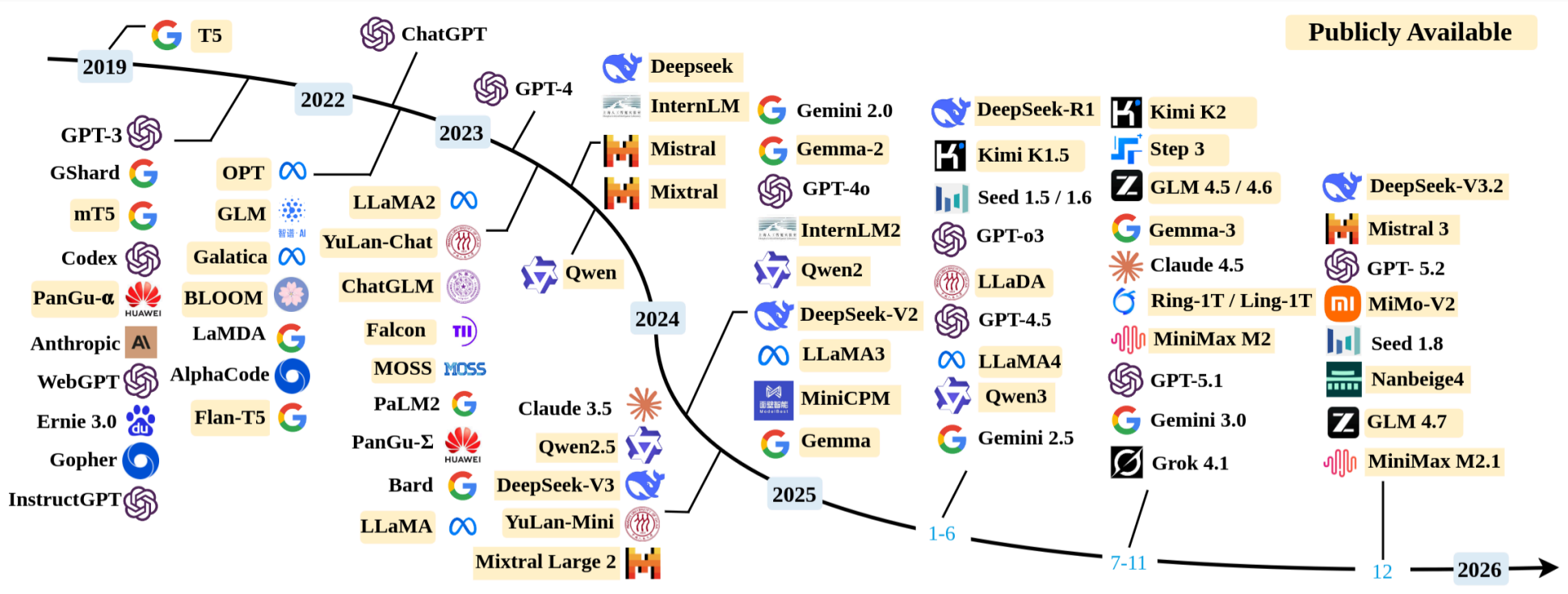


Source: [Claude API Docs](#)

Open-Weight Models

or: Running an LLM on Your Own Computer

A timeline of large language models



Source: Zhao et al. (2026)

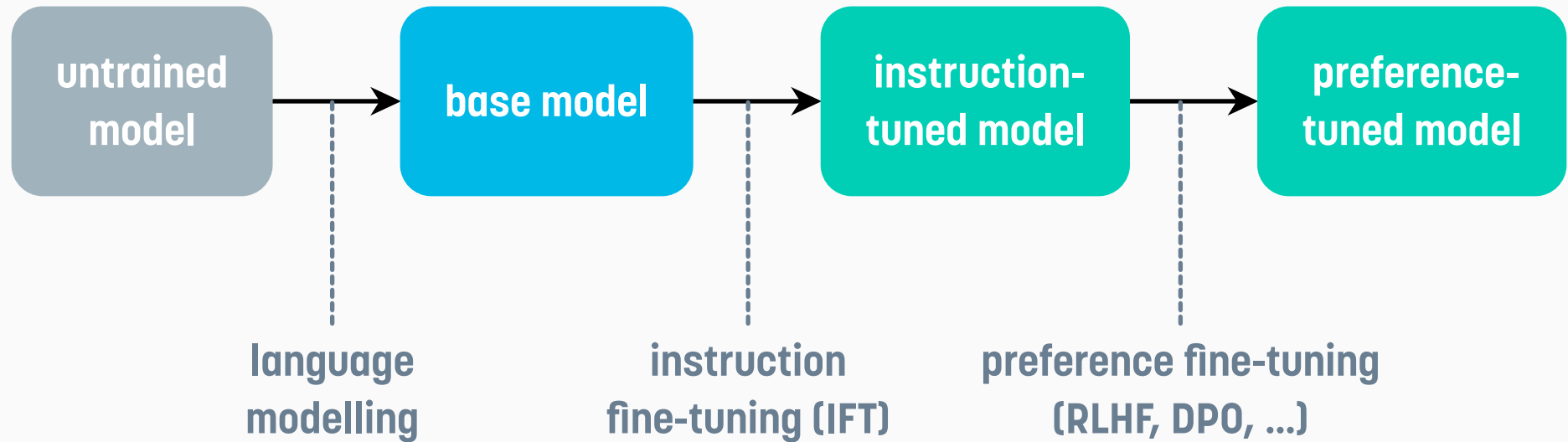
Open-weight models

- Many companies and research labs make trained models **openly available**.
 - [!\[\]\(cd3e54d951a9fb854f48e4697cf550f9_img.jpg\) DeepSeek](#) models (private Chinese company)
 - [!\[\]\(cc729e263f29c0a76fbdc4cfe67fceb0_img.jpg\) Gemma](#) models, by Google Deepmind
 - [!\[\]\(90d36d418f8f7ab67431ba2525e00a5e_img.jpg\) Llama](#) models, by Meta AI
 - [!\[\]\(f70e40faeec369ff477dbaef549ee05b_img.jpg\) Mistral](#) models (private French company)
 - [!\[\]\(ca68c0c79a5dc0026aa1d011fda2b676_img.jpg\) Olmo](#) models, by Ai2 (Allen Institute)
 - [!\[\]\(caba7331972dceb944f99aa56fee2f81_img.jpg\) Qwen](#) models, by Alibaba Cloud
 - ...
- In contrast, **closed models** can only be accessed through a website or an API.
 - e.g. GPT-5, Claude, Grok, Gemini, ...
 - Potential privacy concerns, especially with sensitive data!

Software and tools

-  **LM Studio** provides a GUI to download & run LLMs on your own computer.
 - Suitable for exploration and accessible for less technically versed users
-  **Ollama** is a more lightweight solution to run LLMs as a background service.
 - MacOS & Windows: simple GUI app / Linux: command-line only
 - Exposes an API so LLMs can be called from other apps and scripts
 - can seamlessly exchange a commercial LLM (e.g. ChatGPT) with a local Ollama one!
-  **HuggingFace** is the go-to ecosystem for interacting with LLMs in Python.

Reminder: LLM training pipeline



Know your model!

- **Base models** are pretrained LLMs *without* further alignment.
 - They won't “respond” like an assistant; they are pure next-word predictors.
 - They can be more prone to generating harmful/problematic responses.
- **Instruction fine-tuned models** are more suitable for complex tasks.
 - They will “follow instructions” like an assistant.
 - They may or may not have undergone additional *preference alignment* steps.

💡 Pay attention to the type of model if you want to run LLMs locally!

- Base models will be the default; sometimes they have 'pt' in the name.
- Instruction fine-tuned models often have 'instruct' or 'it' in the name.

The size of large language models

- The size of an LLM is typically expressed in the **number of trainable parameters**.
- Many open-weight LLMs are released in multiple **variants of different sizes**.
 - e.g. [🔗 Gemma 3](#) models:
270M – 1B – 4B – 12B – 27B

Model	Year	Params
BERT	2018	340M
GPT-2	2019	1.5B
GPT-3	2020	175B
GPT-4	2023	(1.7T?)
Qwen3	2025	4B–235B
GLM-5.1	2026	754B

Hardware requirements of large language models

- Most LLMs require a **GPU** with a certain amount of **memory (VRAM)** to run.
 - Llama-2 7B: 10GB VRAM (e.g. *many modern gaming GPUs*)
 - Llama-2 13B: 24GB VRAM (e.g. *high-end gaming GPUs*)
 - Llama-2 70B: 80GB VRAM (e.g. *dedicated supercomputing GPUs*)
- Rule of thumb: **1B–7B** models can usually run on **consumer-grade GPUs**.
- Bigger models can be made to require less memory with **quantization**.
 - Quantization: reducing the floating-point precision of the stored parameters
 - e.g. Gemma 3 27B drops from 54GB (16-bit) to 14GB (4-bit)

Thank you!

